Assignment 1

Mathematics for Cryptography

2025

In this assignment, you are getting familiar with and practise modular computation. Therefore, you should do all (or as much as possible of) the computation by hand. This also forces you to discover simplifications – or "shortcuts" as the Oxford professor Marcus Du Sautoy puts it.[1]

We would like you to explore these problems. The more patterns you discover, interesting observations you make, and questions you find, the more and deeper you learn.

Remember that you should work on all the problems in this assignment, unless it is indicated as optional. Each problem can give you new insights, knowledge, and skills. Problems denoted as *optional* require more time to solve but can give you more insight.

We hope you enjoy this first assignment of the course.

Problem 1

The *Vigenère cipher* is a method of encrypting alphabetic text using a simple form of polyalphabetic substitution. It is named after Blaise de Vigenère (1523–1596), although it was actually developed earlier by Giovan Battista Bellaso (1505–??). The Vigenère cipher was considered a significant advancement in cryptography because it was more resistant to frequency analysis compared to earlier ciphers, such as the Caesar cipher.

The Vigenère cipher uses a keyword or key, which is a sequence of letters. This key is repeated as many times as needed to match the length of the plaintext. For example, if the plaintext is "MATHEMATICSFORCRYPTOGRAPHY" and the key is "HELLOWORLD", the key is repeated to become "HELLOWORLDHELLOWORLDHELLOW". Each letter of the plaintext is encrypted by shifting it forward in the alphabet by a number of positions determined by the corresponding letter in the key. The number of positions is based on the letter's position in the alphabet (A = 0, B = 1, ..., Z = 25). Continuing the example above, the ciphertext is "TEESSIOKTFZJZCQNMGERNVLAVU".

(a) To practise, **encrypt** the plaintext "MATHEMATICSFORCRYPTOGRAPHY" again, but now with the key "LEMON" to get a new ciphertext. Afterwards, explain briefly the **decryption** process using this example.

(b) Give a **mathematical description** of the encryption and decryption process of the Vigenère cipher. Specifically, make it clear how we can work

[1] (Optional) See his book or this particular video of his.

with letters mathematically and provide the mathematical equations for both encryption and decryption using modular arithmetic.

(c) Let us denote the ciphertext the plaintext P, the key K, and the ciphertext C in a Vigenère system. **Analyse** the cases when two of the three are captured by the adversary.

(d) Consider the table below. Think about and **describe** how this table is related to the Vigenère cipher. Give it some time to contemplate and try to find some insights. This is an open question, but make sure that you are happy with and can be proud of your answer.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

(e) (Optional) **Implement** the Vigenère encryption and decryption in Python or Sage.

Problem 2

On one of the slides, we saw that $26! = 403\ 291\ 461\ 126\ 605\ 635\ 584\ 000\ 000$.

(a) Notice that there are six 0s at the end of this number. Think about the reasons why a number ends with one 0, or with multiple zeros, and **explain** why this number ends with exactly six 0s.

(b) **Compute** by hand how many 0s are in the binary representation of 26!. **Describe** how you did this computation.

(c) **Compute** by hand how many 0s are in the hexadecimal representation of 26!. **Describe** how you did this computation.

(d) (Optional) Generalise the results above, that is, **determine** the number of trailing zeros of $n!$ (where n is a positive whole number) in the decimal, the binary, the hexadecimal systems. We can denote these numbers as $T_{10}(n), T_2(n), T_{16}(n)$, respectively.

We expect one of two kinds of solutions: a mathematical formula or a Python program. In both cases **explain** how you found this solution and what the mathematical reasoning behind it.

Problem 3

The Enigma machine was a cipher device used by the military of Nazi Germany during the 1930s and 1940s. It employed a complex substitution cipher that allowed for a vast number of possible combinations, making it a formidable tool for secure communication. Your task is to analyse the Enigma machine's operation, calculate the number of combinations, and understand its vulnerabilities.

(a) Watch the video "How did the Enigma Machine work?" by Jared Owen, `https://www.youtube.com/watch?v=ybkkiGtJmkM`. **Summarise** the most important components of the Enigma machine and **explain** how it encrypts and decrypts messages including the role of the rotors and the plugboard. Please provide a concise description of **at most 180 words** that focuses on the process from when a letter is pressed to when the corresponding light illuminates.

(b) In the Numberphile video "158,962,555,217,826,360,000 (Enigma Machine)" `https://www.youtube.com/watch?v=G2_Q9FoD-oQ`, James Grimes presents a formula for calculating the number of combinations for the Enigma machine:

$$\binom{5}{3} 3! \cdot 26^3 \cdot \frac{26!}{6! \cdot 10! \cdot 2^{10}}.$$

Try to understand the James's explanation and explain each term in this formula using your own logic and words. **Explain** your understanding including what each component represents in the context of the Enigma machine?

(c) **Calculate** what the number of settings is in the Enigma machine if there were n rotors from which $k \leq n$ rotors must be used?

(d) **Calculate** the total number of combinations for the plugboard when using up to only 2 plugs, instead of the 10 plugs shown in the video. (Here, we focus only on the plugboard combinations. The rest of the formula can be neglected.)

(e) Watch the video "Flaw in the Enigma Code" by Numberphile, `https://www.youtube.com/watch?v=V4V2bpZlqx8`. In this assignment, you will analyse the implications of a specific property of substitution ciphers using a smaller example. Consider a simplified alphabet consisting of 4 letters: A, B, C, and D. In this context, analyse how the restriction of not allowing a letter to map to itself affects the total number of valid mappings in a substitution cipher:

Calculate the total number of unrestricted mappings for the 4 letters. Then, **compute** the number of valid mappings where no letter maps to itself. Finally, think about and **describe** the implications of the reduction in valid mappings on the security of a substitution cipher using this 4-letter alphabet.

Problem 4

Consider the set of all integers modulo 24: $\{0, 1, 2, 3, 4, \ldots, 21, 22, 23\}$. This set is often denoted as $\mathbb{Z}/24\mathbb{Z}$ or $\mathbb{Z}_{24}$ in mathematics and in cryptography. In this problem, we will explore how multiplication works in $\mathbb{Z}_{24}$. (At this point, it's a good idea for you to stop and **experiment** with multiplication modulo 24 within this set. You can try, for instance, to compute $2 \cdot 21, 3 \cdot 22, 4 \cdot 6, 20 \cdot 6, 7 \cdot 7$ and record as much of your observation as possible. Moreover, you could also try to explain any interesting phenomena you notice.)

(a) **Find** all integer $k \in \mathbb{Z}_{24}$ such that $\gcd(k, 24) = 1$. Having found these numbers, **determine** $\varphi(24)$.

(b) The integers that you found above are called *units*. The set of all units modulo 24 is denoted by $\mathbb{Z}/24\mathbb{Z}^*$ or $\mathbb{Z}_{24}^*$.

Create a multiplication table of all units modulo 24 (i.e. all elements in $\mathbb{Z}_{24}^*$).

(c) **Find** the inverse of each element of $\mathbb{Z}_{24}^*$.

(d) **Determine** the following sum and product:

$$\sum_{k \in \mathbb{Z}_{24}^*} k \bmod 24 \quad \text{and} \quad \prod_{k \in \mathbb{Z}_{24}^*} k \bmod 24.$$

What do you observe? Try to **reason** about your observations.

(e) 24 is not a prime; actually it is a highly composite number with many small prime factors because of its prime factorisation: $24 = 2^3 \cdot 3$. Therefore, it is interesting to examine other kinds of numbers with simpler structures, like prime numbers and products of two primes. These numbers have more units in relation to the number, that is, $\frac{\varphi(n)}{n}$ is closer to 1.

Compute the following expressions and **describe** your observations or questions.

$$\frac{\varphi(11)}{11}, \quad \sum_{k \in \mathbb{Z}_{11}^*} k \bmod 11, \quad \text{and} \quad \prod_{k \in \mathbb{Z}_{11}^*} k \bmod 11,$$

$$\frac{\varphi(15)}{15}, \quad \sum_{k \in \mathbb{Z}_{15}^*} k \bmod 15, \quad \text{and} \quad \prod_{k \in \mathbb{Z}_{15}^*} k \bmod 15.$$